

1 Solution — GIAM 6.5

Problem 2

1.1 Problem

Find a bijection from the set of odd squares,

$$S = \{1, 9, 25, 49, \dots\},$$

to the non-negative integers,

$$\mathbb{Z}^{\text{noneg}} = \{0, 1, 2, 3, \dots\}.$$

Prove the function is both injective and surjective, and find its inverse.

1.2 Setup

The non-negative odd integers are $1, 3, 5, 7, \dots$ — i.e. $\{2k + 1 : k \in \mathbb{Z}^{\text{noneg}}\}$. Their squares form the set of odd squares,

$$S = \{(2k + 1)^2 : k \in \mathbb{Z}^{\text{noneg}}\} = \{1, 9, 25, 49, 81, 121, \dots\}.$$

This *parametrization* is itself the bijection: every element of S arises from exactly one $k \in \mathbb{Z}^{\text{noneg}}$.

1.3 The Bijection

Define

$$f : S \longrightarrow \mathbb{Z}^{\text{noneg}}, \quad f(n) = \frac{\sqrt{n} - 1}{2}.$$

Equivalently, $f((2k + 1)^2) = k$: the formula reads each odd square as $(2k + 1)^2$, takes its positive square root $2k + 1$, and recovers the index k .

A few values:

n	$\sqrt{n}$	$f(n)$
1	1	0
9	3	1
25	5	2
49	7	3
81	9	4

Table 1.1

Well-defined: every element of S has the form $(2k+1)^2$ for a unique $k \in \mathbb{Z}^{\text{noneg}}$, and $f((2k+1)^2) = k \in \mathbb{Z}^{\text{noneg}}$. So f does map S into $\mathbb{Z}^{\text{noneg}}$.

1.4 Injectivity

Suppose $f(m) = f(n)$ for some $m, n \in S$. Write $m = (2j+1)^2$ and $n = (2k+1)^2$ with $j, k \in \mathbb{Z}^{\text{noneg}}$. Then

$$j = f(m) = f(n) = k,$$

so $2j+1 = 2k+1$, hence

$$m = (2j+1)^2 = (2k+1)^2 = n.$$

Thus f is injective.

1.5 Surjectivity

Take any $k \in \mathbb{Z}^{\text{noneg}}$. The integer $2k+1$ is a non-negative odd integer, so $(2k+1)^2 \in S$. Computing,

$$f((2k+1)^2) = \frac{(2k+1) - 1}{2} = \frac{2k}{2} = k.$$

So every $k \in \mathbb{Z}^{\text{noneg}}$ lies in the image of f ; that is, f is surjective.

Together with injectivity, this makes f a **bijection**. ■

1.6 The Inverse

Since f is a bijection, it has a (unique) two-sided inverse $f^{-1} : \mathbb{Z}^{\text{noneg}} \rightarrow S$. The natural candidate, undoing the steps of f , is

$$f^{-1}(k) = (2k + 1)^2.$$

A few values:

k	$f^{-1}(k)$
0	1
1	9
2	25
3	49
4	81

Table 1.2

To confirm this really is f^{-1} , check both compositions:

$$(f \circ f^{-1})(k) = f((2k + 1)^2) = \frac{(2k + 1)^2 - 1}{2} = k.$$

For the other direction, take any $n = (2k + 1)^2 \in S$:

$$(f^{-1} \circ f)(n) = f^{-1}(n) = (2k + 1)^2 = n.$$

So $f \circ f^{-1} = \text{id}_{\mathbb{Z}^{\text{noneg}}}$ and $f^{-1} \circ f = \text{id}_S$, confirming $f^{-1}(k) = (2k + 1)^2$ is the inverse.

1.7 Remark — Sequences as Bijections

The bijection f amounts to saying: the k -th non-negative integer corresponds to the k -th odd square (zero-indexed). This is a special case of a much more general pattern.

Any *enumeration* of a set A — that is, a list $(a_0, a_1, a_2, \dots)$ of distinct elements that exhausts A — defines a bijection

$$\varphi : \mathbb{Z}^{\text{noneg}} \rightarrow A, \quad \varphi(k) = a_k.$$

For our problem, $a_k = (2k+1)^2$, so $\varphi = f^{-1}$ and f is the inverse “find the index” map. The same recipe shows that *any* countably infinite set is in bijection with $\mathbb{Z}^{\text{noneg}}$ — that is the meaning of “countably infinite.”